# Setup Guide – Composite Behavioral Modeling for Identity Theft Detection

---

## 1. System Requirements

Before running the project, ensure your system meets the following:

- **Operating System:** Windows / Linux / macOS
- **Python Version:** 3.10 or higher
- **Framework:** Django 3.2
- **Database:** MySQL (via XAMPP or standalone)
- **Browser:** Chrome / Edge / Firefox

---

## 2. Installation Commands

Open **Command Prompt / Terminal** and run:

```
# 1. Upgrade pip
python -m pip install --upgrade pip

# 2. Install Django
pip install django==3.2

# 3. Install Required Libraries
pip install pandas scikit-learn joblib

# 4. Install MySQL Connector
pip install pymysql
```

---

## 3. Project Setup

### Step 1: Extract Project Files

- Extract the project folder to your desired location
- Open the folder in **VS Code or Terminal**

---

### Step 2: Configure Database

1. Open **XAMPP Control Panel**
2. Start:
   - ☑ MySQL
3. Open **phpMyAdmin**
4. Create a new database:

```
identity_theft_detection
```

---

### Step 3: Update `settings.py`

Set your database configuration:

```
DATABASES = {
    'default': {
        'ENGINE': 'django.db.backends.mysql',
        'NAME': 'identity_theft_detection',
        'USER': 'root',
        'PASSWORD': '',
        'HOST': '127.0.0.1',
        'PORT': '3306',
    }
}
```

---

# 4. Execution Workflow

### 🔹 Step A: Apply Migrations

```
python manage.py makemigrations
python manage.py migrate
```

---

### 🔹 Step B: Run the Server

```
python manage.py runserver
```

---

### 🔹 Step C: Server Startup

- Wait until you see:

```
Starting development server at http://127.0.0.1:8000/
```

---

# 5. Using the Application

1. Open browser and go to:

```
http://127.0.0.1:8000/
```

2. Register a new user
3. Login to the system
4. Navigate to:

```
Predict Identity Theft Detection
```

5. Enter required input details
6. Click:

```
Predict
```

7. View prediction result:

- Fraud Detected / Not Fraud

---

# 6. Project Architecture

- **Frontend:** HTML, CSS
- **Backend:** Django 3.2
- **Database:** MySQL
- **Machine Learning Models:**
  - Logistic Regression
  - Support Vector Machine (SVM)
  - Naive Bayes
- **Processing:**
  - Data preprocessing
  - Feature encoding
  - Model prediction

---

# 7. Important Notes

- Ensure MySQL service is running before starting the project
- Input data must follow expected format for accurate prediction
- Model performance depends on proper preprocessing

---

# 8. Troubleshooting

### ❌ Error: Database connection failed

- ✔ Check MySQL is running
- ✔ Verify credentials in `settings.py`

---

# 9. Conclusion

This system enables detection of identity theft behavior using machine learning models integrated with a web-based interface. Proper setup and configuration ensure smooth execution and accurate predictions.